B. APPLICATION OF GIRSANOV'S THEOREM TO HESTON'S MODEL

Under the real world probability measure, $\mathbb{P}$, the asset dynamics of the Heston model are given by equations (18), (19), and (20). Quoted again these are:

$$\begin{aligned} \frac{dS}{S} &= \mu dt + \sqrt{v} S dX_1^{\mathbb{P}}, \\ dv &= \kappa\left(\theta - v\right) dt + \xi\sqrt{v} dX_2^{\mathbb{P}}, \\ \rho dt &= dX_1^{\mathbb{P}} dX_2^{\mathbb{P}}. \end{aligned}$$

Define the market prices of risk (or Girsanov kernel, which describes the adapted stochastic process used in Girsanov's theorem to shift the drift of a Brownian motion when changing from the real world probability measure to the risk-neutral probability measure) as:

$$\alpha_S = \frac{\left(\mu^{\mathbb{P}} - r\right) S}{\sqrt{v} S} = \frac{\mu^{\mathbb{P}} - r}{\sqrt{v}}, \qquad \alpha_v = \frac{\lambda v}{\xi^{\mathbb{P}}\sqrt{v}} = \frac{\lambda}{\xi^{\mathbb{P}}}\sqrt{v}, \tag{42}$$

where λ is a parameter. This parameter is known as the variance risk premium. In other words, this represents the extra return an investor demands for being exposed to unexpected movements in the variance process. Note that the term $\mu^{\mathbb{P}} - r$ denotes the risk premium. In other words, this represents the extra return an investor requires for taking on one unit of risk from the shocks that drive the asset price. A risk-averse agent will only hold that exposure if they are compensated for the uncertainty it introduces, so this term captures the reward they demand for bearing that underlying source of risk. The market prices of risk, (42), are Sharpe ratios, they divide the risk premium by the corresponding instantaneous volatilities. The two-dimensional Girsanov Theorem yields:

$$\begin{aligned} \left.\frac{d\mathbb{Q}}{d\mathbb{P}}\right|_{\mathcal{F}_t} = \exp\Bigg(&- \int_0^t \frac{\mu^{\mathbb{P}} - r}{\sqrt{v}}\, dX_S^{\mathbb{P}} - \int_0^t \frac{\lambda}{\xi^{\mathbb{P}}}\sqrt{v}\, dX_v^{\mathbb{P}} \\ &+ \int_0^t \frac{\left(\mu^{\mathbb{P}} - r\right)\lambda\rho^{\mathbb{P}}}{\xi}\, ds - \frac{1}{2}\int_0^t \left(\frac{\left(\mu^{\mathbb{P}} - r\right)^2}{v} + \frac{\lambda^2 v}{\xi^2}\right) ds\Bigg), \end{aligned} \tag{43}$$

such that:

$$\begin{aligned} dX_S^{\mathbb{Q}} &= dX_S^{\mathbb{P}} + \alpha_S dt, \\ dX_v^{\mathbb{Q}} &= dX_v^{\mathbb{P}} + \alpha_v dt, \end{aligned}$$

are increments of standard Brownian motions under the risk-neutral probability measure, $\mathbb{Q}$. To keep consistency with notation, let $dX_S = dX_1$ and $dX_v = dX_2$. Note that $dX_1^{\mathbb{Q}} dX_2^{\mathbb{Q}} = \rho^{\mathbb{P}} dt$. Thus, when changing the probability measure, the correlation coefficient remains the same, $\rho^{\mathbb{Q}} = \rho^{\mathbb{P}}$. Note that the stochastic discount factor is $M_t = e^{-rt}\frac{d\mathbb{Q}}{d\mathbb{P}}\big|_{\mathcal{F}_t}$. Thus, under the risk-neutral probability measure, $\mathbb{Q}$, the asset dynamics of the Heston model are given by:

$$\begin{aligned} \frac{dS}{S} &= r dt + \sqrt{v} S dX_1^{\mathbb{Q}}, \\ dv &= \kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - v) dt + \xi\sqrt{v} dX_2^{\mathbb{Q}} - \lambda v dt \\ &= \left(\kappa^{\mathbb{P}}\theta^{\mathbb{P}} - \left(\kappa^{\mathbb{P}} + \lambda\right) v\right) dt + \xi\sqrt{v} dX_2^{\mathbb{Q}} \\ &= \kappa^{\mathbb{Q}}\left(\theta^{\mathbb{Q}} - v\right) dt + \xi\sqrt{v} dX_2^{\mathbb{Q}}, \\ \rho dt &= dX_1^{\mathbb{Q}} dX_2^{\mathbb{Q}}. \end{aligned}$$

The speed of mean reversion and the long-term mean are:

$$\kappa^{\mathbb{Q}} = \kappa^{\mathbb{P}} + \lambda, \qquad \theta^{\mathbb{Q}} = \frac{\kappa^{\mathbb{P}}\theta^{\mathbb{P}}}{\kappa^{\mathbb{P}} + \lambda}.$$

Note that, $\kappa^{\mathbb{P}}\theta^{\mathbb{P}} = \kappa^{\mathbb{Q}}\theta^{\mathbb{Q}}$.